



GUJRAT TECHNOLOGICAL UNIVERSITY

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Electronics and communication Department



INDUSTRIAL ROBOTS: THE ROLE OF SAFETY STANDARDS



1. BUREAU OF INDIAN STANDARDS (BIS)

India's National Standards Body

BIS develops Indian Standards for products, processes and services.

Its main goals are:

Safety • Quality • Reliability • Performance • Sustainability

BIS works in:

Standards • Testing • Certification • Conformity Assessment •

Consumer Protection

2. WHAT IS A STANDARD?



A standard is a set of **technical requirements and guidelines** that helps make a product or system:

Safe • Reliable • Consistent • Fit for Use

3. WHY DO ENGINEERING STANDARDS

Modern engineering systems contain many interconnected parts

A failure in one part can affect the **whole system**.

Standards help engineers achieve:

BIS in Engineering

Sensors | Transducers | Batteries | Machines | Electronics | IoT Systems | Robots

4. INDUSTRIAL ROBOTS — AN ENGINEERING EXAMPLE

POWERFUL AUTOMATION NEEDS STRONG SAFETY

An industrial robot is a programmable machine used to perform manufacturing tasks automatically.



Common Applications:



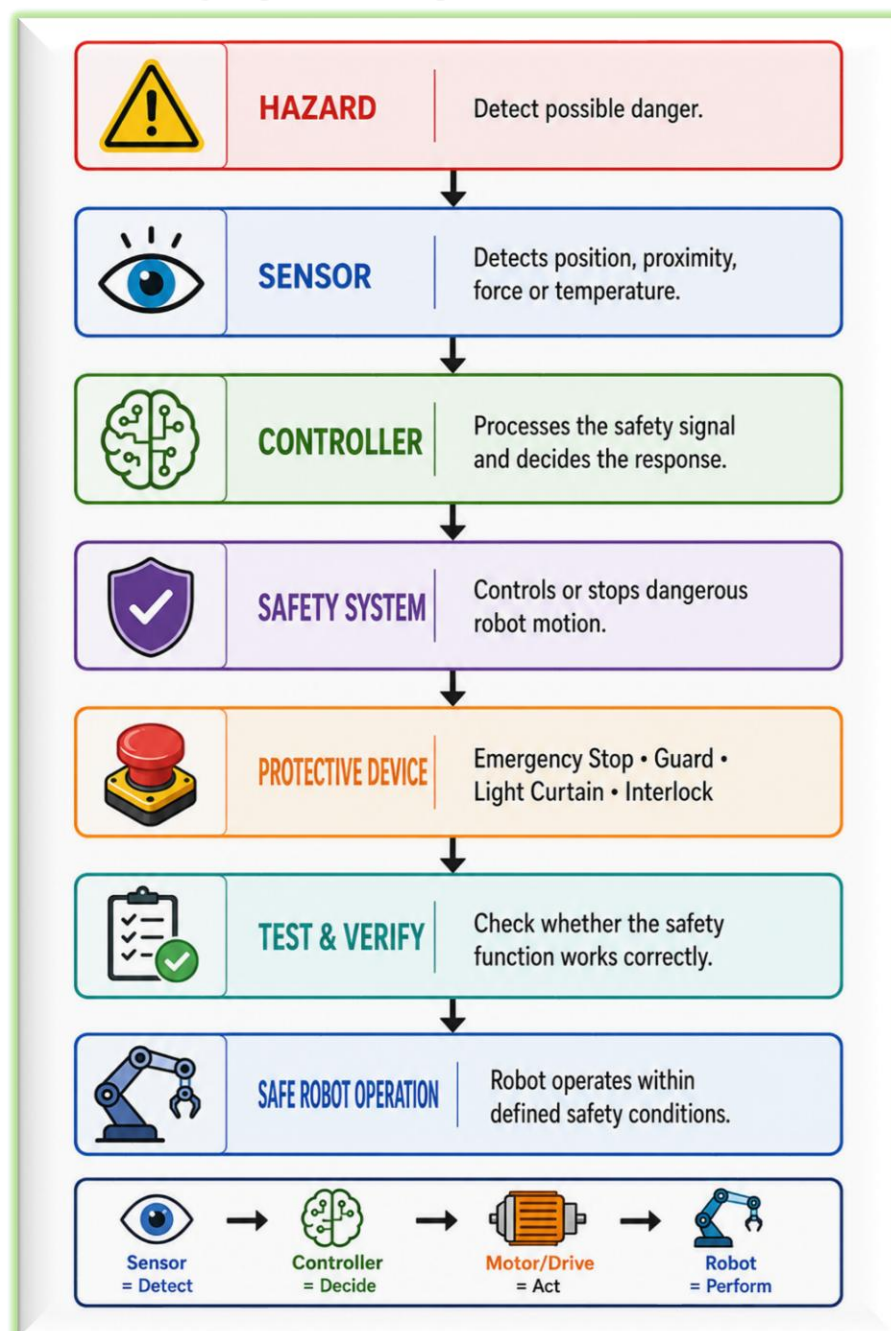
Robots provide:



But automatic machines can also create safety hazards.

5. ROBOT SAFETY: FROM SENSING TO SAFE OPERATION

Industrial robots can move **fast, automatically and with high force**, creating risks such as collision, crushing, impact and unexpected movement.

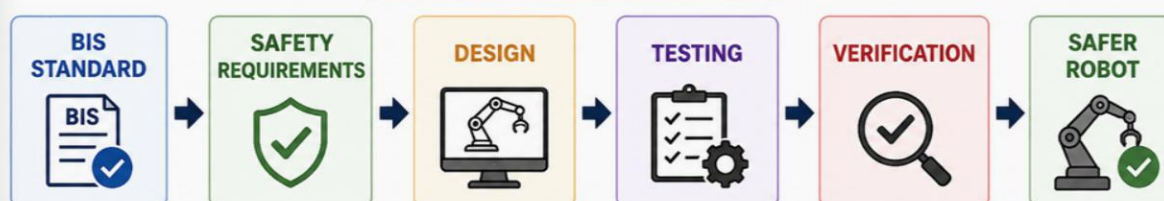


6. BIS + ECE + INDUSTRIAL ROBOTS

Industrial robots combine sensors, controllers and actuators for automatic operation. BIS standards help define safety requirements, testing and verification for safer robot systems.



Example: Industrial Robots



ECE Connection



Sensors detect. Controllers decide. Actuators act.



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Explore the Technology, Standards & Safety Behind Industrial Robots

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